

Supplementary material

Learning what to inspect: Task-driven multimodal C-scan acquisition for compression-after-impact assessment

Supplement to the offline validation study on 50 physical specimens, 48 capture groups and six domains.

S1 Data provenance and reproducibility

Surface and internal images come from six version-1 post-impact datasets; the CAI-strength source is the version-3 compression-after-impact dataset. Original records are published under CC BY 4.0. The study’s registered crops, overlays and feature representations are derived assets, and their release requires correct attribution and author confirmation of the intended release scope. Source version and specimen correspondence are recorded independently of filenames; a layer count is not used to infer a per-specimen thickness. In particular, photographs taken after compression in the CAI dataset are not substituted for the post-impact surface input.

The source authors are Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki and Shin-ichi Takeda. The image and mechanical datasets are described in their Data in Brief articles [1, 2]. Table S1 maps the exact dataset versions. The local cohort contains 276 specimens in 259 source-related capture groups, split as 161/152 TRAIN, 50/48 VALID and 65/59 reserved TEST physical specimens/groups. The reserved TEST partition was not evaluated.

Domain	Configuration verified from source record	Version / DOI
74t7kcdgkr	8-layer cross-ply	v1, 10.17632/74t7kcdgkr.1 [3]
yfyg8jm46	16-layer cross-ply	v1, 10.17632/yfyg8jm46.1 [4]
xcmzfsbd9t	24-layer cross-ply	v1, 10.17632/xcmzfsbd9t.1 [5]
ykhs7s2dck	8-layer quasi-isotropic	v1, 10.17632/ykhs7s2dck.1 [6]
w68dtmpfyf	16-layer quasi-isotropic	v1, 10.17632/w68dtmpfyf.1 [7]
cgtmjyggtm	24-layer quasi-isotropic	v1, 10.17632/cgtmjyggtm.1 [8]
CAI measurement source	Compression-after-impact strength workbook	v3, 10.17632/8scdmfdecb.3 [9]

Table S1. Source mapping. The six image domains use impacted-surface views and registered internal-image crops. Labels use the author workbook’s CAI_STRENGTH field in MPa, not impact energy, delamination area or normalized strength reduction.

The predictor uses frozen 64×512 surface and internal descriptors, 64-dimensional projections and cell representations, and all-cell plus measured-cell pooling. The spatial actor uses 128-dimensional tokens, two encoder layers, four attention heads, a 256-dimensional feed-forward block and zero actor attention dropout. The query and 64 cells are processed together. The mean-feedback actor has 91,650 parameters, compared with 378,978 for each spatial actor and 64 for learned-static. No capacity-matched interpretation is intended.

Predictor training item	Saved setting
Supervision	CAI strength in MPa; mean Huber loss of $(\hat{y} - y)/s$ against zero, delta 1
Residual and output normalization	Fit-target mean; $s = \max(\text{std}(y_{\text{fit}}), 1 \text{ MPa})$; OOF constants fitted separately
Batch / specimen sampling	32; uniform domain, then uniform physical specimen within domain
Mask sampling probabilities	10% zero; 60% 1–16 cells; 20% 17–48 cells; 10% full 64

Predictor training item	Saved setting
Partial-mask routes	1–16: Random/Center-first/Geometry-spread/Serpentine prefixes; 17–48: random permutation
Optimizer / learning rate / weight decay	AdamW / 0.0003 / 0.0001
Gradient-norm clipping	1
Maximum updates / validation interval / patience	2000 / 250 / 4; checkpoint improvement tolerance 10^{-12}
Common candidate seeds	MEAN_SC 2026091201; SPATIAL_SC 2026091202; SPATIAL_C 2026091203
Common selected / completed updates	MEAN_SC 1750/2000; SPATIAL_SC 750/1750; SPATIAL_C 500/1500
Validation library	Zero state and affordable prefixes of four fixed routes on the same 50 VALID specimens; budget 0.25
Selection measure	Route mean within specimen, specimen mean within domain, equal mean over six domains of A
Candidate choice	Minimum eligible validation area; ties within 10^{-8} use fewer parameters, then name
OOF seeds / selected updates	2026091211–2026091213 / 1250, 750, 1000
OOF completed updates	2000, 1750, 2000
OOF fit / query physical counts	104/57, 108/53, 110/51; no fit/query capture-group overlap

Predictor-specific settings. Mask probabilities describe independent random draws, not enforced batch counts. Each source-related training group is assigned to one of three folds within its domain, with groups kept intact. A query specimen uses the predictor fitted outside its fold. The validation library is shared for checkpoint selection, including OOF models; group exclusion from fitting does not imply independent checkpoint validation. Fixed-route validation prefixes are distinct from the later actor trajectories.

Eligibility required finite metrics; full-input MAE below the fit-median constant predictor, full-input MSE below the fit-mean constant predictor, and at least 2% lower full-input MAE than zero-input MAE. Center-first and Geometry-spread endpoint MAEs also had to improve on zero-input MAE. Constants are computed from fitting labels and evaluated on VALID. All three common candidates satisfied these checks before their validation areas were compared.

Native cell costs and the legal acquisition test use float64 arithmetic with tolerance 10^{-12} at budget 0.25. The feature-model cost channel is float32, while legal decisions retain native float64 costs. After an action, cost is the sum of measured-cell costs and history at that cell is its one-based acquisition step divided by 64. Cell identifiers themselves are zero-based, from 0 to 63. The logical X_{obs} buffer is zero in unobserved cells; model-specific masking prevents the environment’s stored full array from becoming visible input.

Actor training item	Saved setting
Optimizer / batch	AdamW / 16
Learning rate / weight decay	0.0003 / 0.0001
Gradient clipping	Norm 1
Discount / critic weight	1 / 0.5
Entropy weight	Linear 0.01 to 0
Budget / terminal-error weight	0.25 / 0.25
Validation interval / patience	250 updates / 4
Selection tolerance	10^{-12}
Main, no-VLM, open-loop, static, mean seeds	2026091301, 2026091302, 2026091303, 2026091304, 2026091305

Actor training item	Saved setting
Corresponding selected updates	250, 1000, 750, 250, 250
Corresponding completed updates	1250, 1250, 1250, 750, 1250
Common predictor / out-of-fold selected updates	1750 / 1250, 750, 1000
Saved research environment	Python 3.13.13; NumPy 2.5.1; PyTorch 2.12.1+cu130; CUDA 13.0

Reproducibility settings. Software versions above are from the saved research protocol, not a claim that manuscript rendering executed the models. Training counts refer to the completed experiment.

Each separately cropped image cell is converted to luminance using RGB coefficients (0.299, 0.587, 0.114), resized to 224×224 with bilinear interpolation (antialiasing enabled, align_corners disabled), replicated across three channels and normalized using ImageNet means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225). The frozen encoder returns one 512-dimensional descriptor per cell. This preprocessing is applied after cell cropping, so it does not mix neighbouring internal cells.

The exact Qwen2.5-VL-7B-Instruct revision is cc594898137f460bfe9f0759e9844b3ce807cfb5. Surface cue confidence is ordinal. A no-reliable-cue response differs from an unavailable response; the fit-side cache has 205 available and six unavailable records. The highest medium/high level restricts only the first legal proposal if affordable, and numerical prior features remain afterwards. The VLM and ResNet sources are attributed separately [10, 11]; the implemented architecture and visibility rules are documented in the main text.

S2 Timing identity and aggregation

Let $c_0=0$, let c_t increase strictly with completed acquisitions, and let $c_T \leq B$. Define e_t as absolute error after acquisition t . Direct normalized left-constant integration is

$$A(B) = \frac{1}{B} \left[\sum_{t=1}^T (c_t - c_{t-1}) e_{t-1} + (B - c_T) e_T \right].$$

Because $e_t = e_0 - \sum_{j=1}^t \delta_j$, substituting the cumulative error changes into the integral gives

$$A(B) = e_0 - \frac{1}{B} \sum_{j=1}^T (B - c_j) \delta_j = e_0 - \sum_{j=1}^T g_j.$$

Each δ_j remains in the held error for the interval from c_j to B , which gives its coefficient. This derivation includes the final holding interval even when $c_T < B$. At $c_j = B$ the area coefficient is zero; J additionally includes $0.25e_T$, so an endpoint observation can still change J . Negative δ_j values are not clipped. The formula also holds for an empty trajectory with $A=e_0$.

Completion stages are right-closed: $(0, 0.0625]$, $(0.0625, 0.125]$, $(0.125, 0.1875]$ and $(0.1875, 0.25]$. Episode contributions are summed within a stage before Random repeats are averaged within a physical specimen. Specimens are then averaged within domains, followed by equal weighting of the six domains. This avoids giving Random’s five runs or longer action sequences more specimen weight. For two methods with unequal initial errors, the contrast is

$$A_{control} - A_{main} = (e_{0,control} - e_{0,main}) + \sum_t g_{t,main} - \sum_t g_{t,control}.$$

All nine saved methods have the same domain-equal initial error, 57.048718 MPa. The identity was checked on all 650 episodes and reproduced saved method areas and paired differences. Maximum episode discrepancy was 2.8422×10^{-14} MPa and maximum method-area discrepancy was 7.1054×10^{-15} MPa. These checks verify arithmetic and aggregation; they do not create independent evidence for a component’s causal effect. The complete 36-row contribution table and machine-readable identity checks accompany the manuscript.

S3 Complete same-cap results

All reported caps query prefixes of trajectories generated with total budget 0.25, using actor remaining budget $0.25 - c_t$. No separate policy is trained or rerun for a smaller displayed cap. Cohort-level MAE pools physical-specimen losses after within-specimen repeat averaging.

Table S2 retains the original five-cap estimates and actual mean and range of unique native-pixel acquisition fractions for every method. Areas are provided once per method in the main table; they do not vary with a queried cap. The complete-input row is separate at fraction 1.0, without an area. Machine-readable source data retain unrounded values, native-pixel counts, unused budgets and aggregation labels.

Table S2 — Center-first

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	47.885	64.882	0.581	0.062350	0.061399–0.062407
0.125	48.827	65.932	0.568	0.111012	0.109491–0.125000
0.1875	49.094	66.277	0.563	0.172513	0.171900–0.187121
0.25	48.027	65.223	0.577	0.234759	0.234676–0.235577

Table S2 — Geometry-spread

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	50.221	67.804	0.543	0.062018	0.061399–0.062130
0.125	46.189	62.421	0.612	0.124548	0.123164–0.124629
0.1875	46.555	62.347	0.613	0.187336	0.186034–0.187407
0.25	46.910	62.278	0.614	0.249970	0.249269–0.250000

Table S2 — Serpentine

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	49.731	64.721	0.583	0.062379	0.062130–0.062407
0.125	48.773	64.778	0.583	0.124592	0.124260–0.124629
0.1875	48.343	63.931	0.593	0.186805	0.186391–0.186851
0.25	48.013	63.041	0.605	0.249184	0.248521–0.249258

Table S2 — Random

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	53.156	69.115	0.525	0.053603	0.046897–0.062500
0.125	49.153	65.038	0.579	0.118768	0.109307–0.125000
0.1875	48.108	63.412	0.600	0.186646	0.171598–0.187500
0.25	46.886	61.941	0.618	0.246943	0.234492–0.250000

Table S2 — Learned-static

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	46.465	62.720	0.609	0.062401	0.062130–0.062500
0.125	48.064	63.994	0.593	0.124803	0.124260–0.125000
0.1875	46.784	62.570	0.611	0.187400	0.187130–0.187500
0.25	46.993	62.168	0.616	0.235297	0.234674–0.249991

Table S2 — No-VLM spatial feedback

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	43.324	60.878	0.631	0.057700	0.046897–0.062500
0.125	43.164	60.169	0.640	0.119824	0.109120–0.124998
0.1875	42.928	58.388	0.661	0.182555	0.171415–0.187407
0.25	42.385	56.534	0.682	0.247174	0.234492–0.250000

Table S2 — VLM mean feedback

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	46.493	60.980	0.630	0.058037	0.046897–0.062500
0.125	44.248	60.221	0.639	0.119530	0.109120–0.125000
0.1875	43.721	59.628	0.646	0.182330	0.171598–0.187407
0.25	43.611	59.012	0.654	0.245789	0.234492–0.250000

Table S2 — VLM spatial feedback (main)

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	45.517	60.482	0.636	0.060459	0.046897–0.062500
0.125	44.114	59.757	0.645	0.121871	0.109120–0.125000
0.1875	44.685	59.472	0.648	0.184924	0.171711–0.187407

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.25	44.286	58.517	0.659	0.247437	0.234492–0.249999

Table S2 — VLM spatial open-loop

Cap	MAE	RMSE	R ²	Mean fraction	Fraction range
0.0	58.551	75.778	0.429	0.000000	0.000000–0.000000
0.0625	45.592	60.699	0.634	0.059237	0.046713–0.062500
0.125	46.145	62.627	0.610	0.122488	0.109120–0.124998
0.1875	46.107	61.920	0.619	0.185473	0.171711–0.187410
0.25	44.822	59.663	0.646	0.248825	0.234863–0.249816

Complete input at fraction 1.0: MAE 41.690 MPa, RMSE 53.944 MPa, R² 0.711; all specimens have fraction 1.0.

S4 Paired comparisons, domain results and quality targets

Table S3 gives all eight control-minus-main contrasts for full-range area, early area and endpoint quality. Positive values favour the main policy. The four descriptive component contrasts have the existing area intervals displayed in the main text. Additional point-estimate contrasts are not assigned invented intervals. The complete same-cap paired CSV retains 40 MAE comparisons with their saved exploratory intervals.

Control	A difference	Early A difference	MAE difference	RMSE difference
Center-first	3.365	2.178	3.741	6.706
Geometry-spread	2.200	2.237	2.624	3.761
Serpentine	3.801	2.528	3.727	4.524
Random	4.484	4.474	2.601	3.423
Learned-static	2.866	3.303	2.707	3.651
No-VLM spatial feedback	-1.513	-0.872	-1.901	-1.984
VLM mean feedback	0.152	0.747	-0.675	0.495
VLM spatial open-loop	1.391	1.406	0.536	1.145

Table S4 retains each domain’s endpoint MAE and full-range area for all nine strategies. Complete five-cap domain values are provided in the 270-row source CSV. Physical domain counts are 9,9,8,9,8,7 in the domain order of Table 1. Negative domain R² values, when present in the source table, are not discarded or set to zero.

Table S4 — 74t7kcdgkr

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	43.492	44.688
Geometry-spread	32.224	34.289

Method	A (MPa)	MAE at 0.25 (MPa)
Serpentine	38.417	35.987
Random	35.878	33.100
Learned-static	34.899	32.927
No-VLM spatial feedback	31.330	28.655
VLM mean feedback	32.761	31.057
VLM spatial feedback (main)	37.428	36.728
VLM spatial open-loop	39.910	35.748

Table S4 — cgtnjyggtn

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	76.463	73.900
Geometry-spread	81.810	80.507
Serpentine	81.500	79.274
Random	82.443	76.217
Learned-static	80.701	76.691
No-VLM spatial feedback	68.749	64.067
VLM mean feedback	69.855	67.782
VLM spatial feedback (main)	67.872	65.704
VLM spatial open-loop	71.314	68.860

Table S4 — w68dtmpfyf

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	41.027	41.174
Geometry-spread	46.925	45.703
Serpentine	48.262	44.619
Random	48.986	45.679
Learned-static	47.552	42.696
No-VLM spatial feedback	47.030	44.088
VLM mean feedback	49.190	43.494
VLM spatial feedback (main)	46.067	43.780
VLM spatial open-loop	49.400	46.986

Table S4 — xcmzfsbd9t

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	55.546	54.320
Geometry-spread	48.923	44.336
Serpentine	49.111	48.254
Random	55.623	51.453
Learned-static	53.866	52.197
No-VLM spatial feedback	48.163	46.433
VLM mean feedback	51.557	48.905

Method	A (MPa)	MAE at 0.25 (MPa)
VLM spatial feedback (main)	50.910	46.382
VLM spatial open-loop	46.824	48.619

Table S4 — yfxyg8jm46

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	51.894	49.782
Geometry-spread	56.217	54.284
Serpentine	58.816	56.151
Random	53.252	49.240
Learned-static	48.420	50.864
No-VLM spatial feedback	48.059	48.408
VLM mean feedback	48.393	48.002
VLM spatial feedback (main)	48.430	48.246
VLM spatial open-loop	49.327	46.185

Table S4 — ykhs7s2dck

Method	A (MPa)	MAE at 0.25 (MPa)
Center-first	22.429	16.787
Geometry-spread	17.765	16.202
Serpentine	17.362	17.549
Random	21.385	19.718
Learned-static	22.423	20.692
No-VLM spatial feedback	18.250	18.124
VLM mean feedback	19.821	16.983
VLM spatial feedback (main)	19.954	19.822
VLM spatial open-loop	22.231	16.668

The 21 integer targets and 21 source-labelled anchors are retained on both the primary and event grids in the accompanying CSVs. A blank attained cost means the quality was not reached; a blank ratio can instead reflect a zero comparator cost. Negative acquisition reductions are retained. Targets are post hoc empirical levels and do not specify engineering acceptance criteria.

At the geometry endpoint anchor $q=46.909917$ MPa, the primary-grid main/geometry/static earliest caps are 0.0625/0.125/0.0625, giving 50%/0% reductions. On the event grid they are 0.031388065/0.093425651/0.062406858, giving 66.403%/49.704%. The main curve later recrosses above this target. All nine partial methods fail to reach the complete-input reference MAE within the observed range on either grid; no missing 0.25–1.0 segment is filled in.

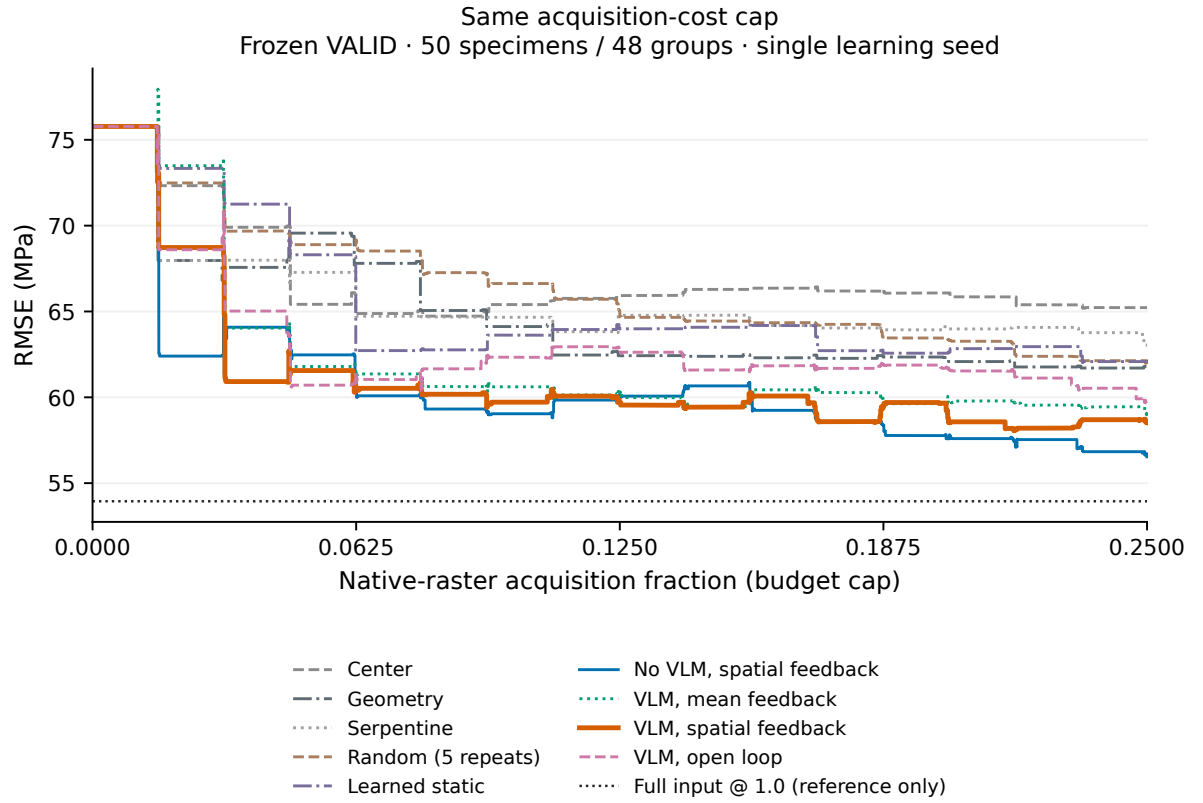


Figure S1: Supplementary RMSE trajectory. Squared errors are averaged over repeats within specimens before pooling and taking the square root. Complete-input reference quality is observed only at fraction 1.0.

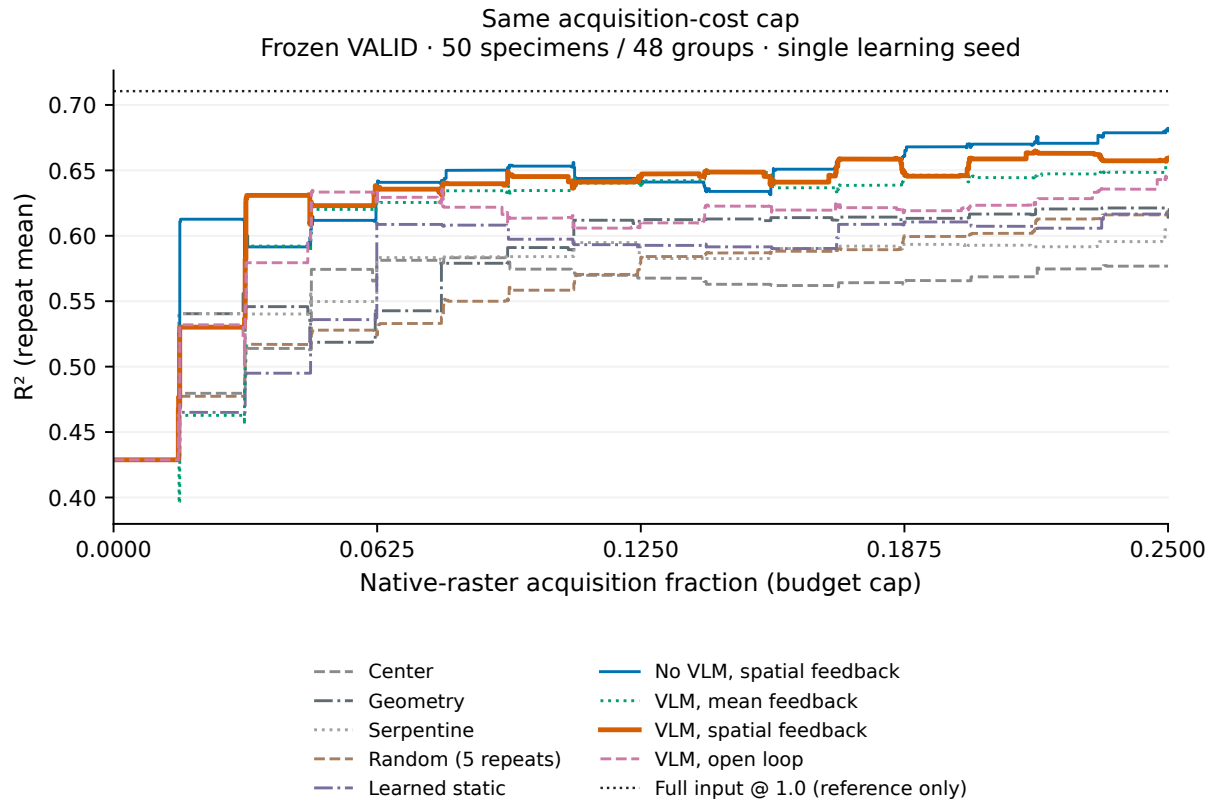


Figure S2: Supplementary R^2 trajectory. Each repeat's specimen-level R^2 is computed before repeat averaging. No prediction ensemble is used.

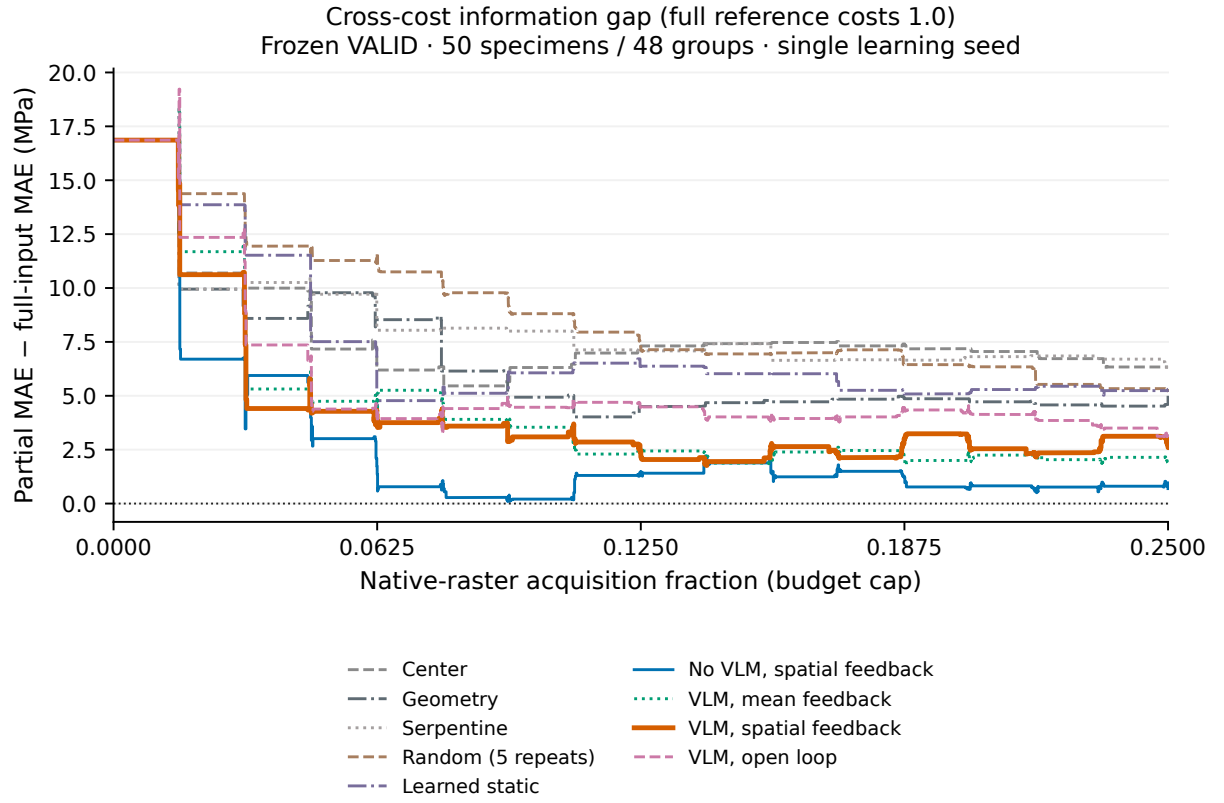


Figure S3: MAE difference from the common predictor’s complete-input reference. Zero is a quality-gap reference, not a prespecified noninferiority margin. No partial method reaches complete-input MAE in the observed range.

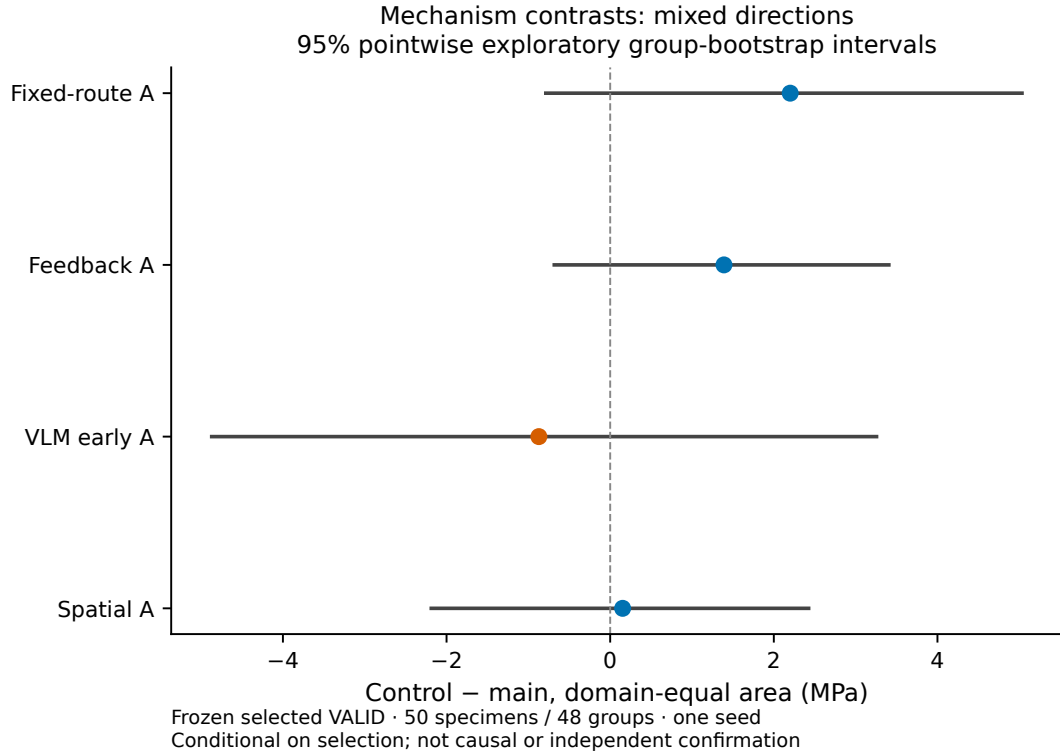


Figure S4: Signed component-related differences and existing 95% pointwise exploratory intervals. Positive differences favour the main policy. The VLM comparison concerns early area; the other contrasts concern full-range area. All four intervals contain zero.

S5 Fixed case illustrations

The three cases were selected before the manuscript timing analysis. Each case below reuses its surface-cue image, final acquired-state overlay and prediction trajectory. The full original seven-image set per case remains in the figure directory with source hashes in the reuse manifest. These are process illustrations, not three independent replications or model-generated explanations. All source specimen imagery derives from the corresponding Hasebe version-1 dataset, licensed CC BY 4.0; registered overlays are copied unchanged. No damage-mask ground truth or language reasoning was added.

VALID surface + frozen VLM cues

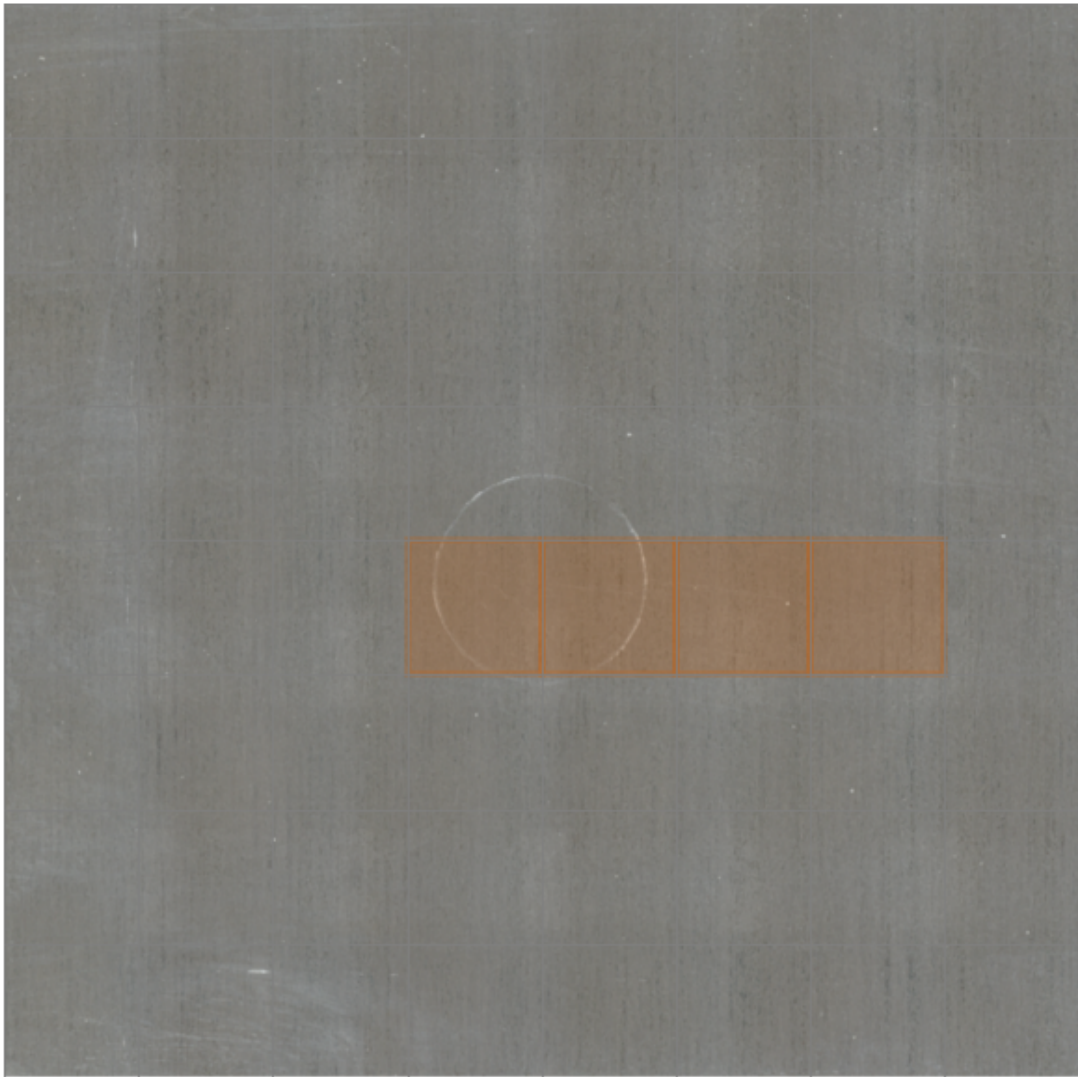


Figure S5: 74t7kcdgkr: c8-16 — surface cues. Actual saved inputs, observations and predictions; unchanged source overlay.

VALID k=16, cost=0.2496
budget endpoint; gray = unobserved

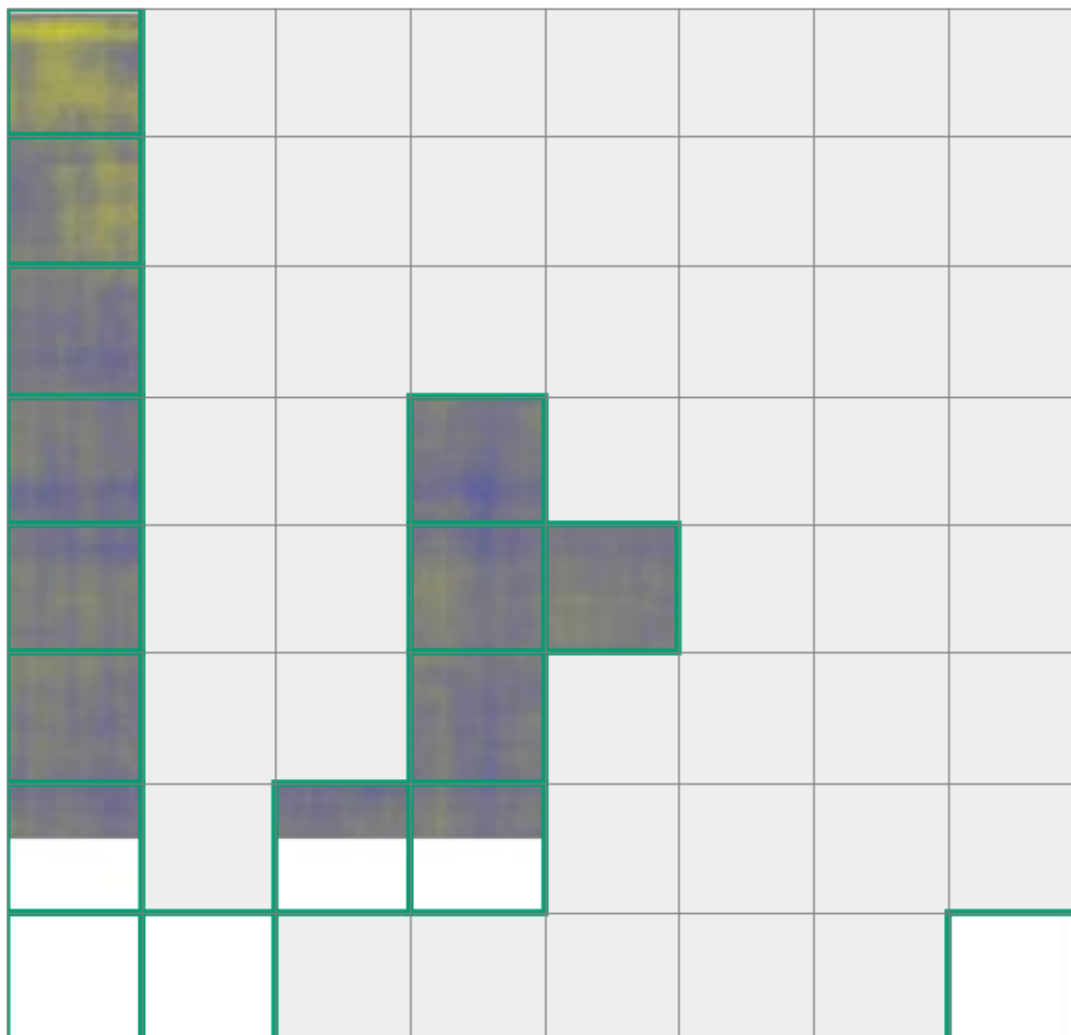


Figure S6: 74t7kcdgkr: c8-16 — measured 16. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

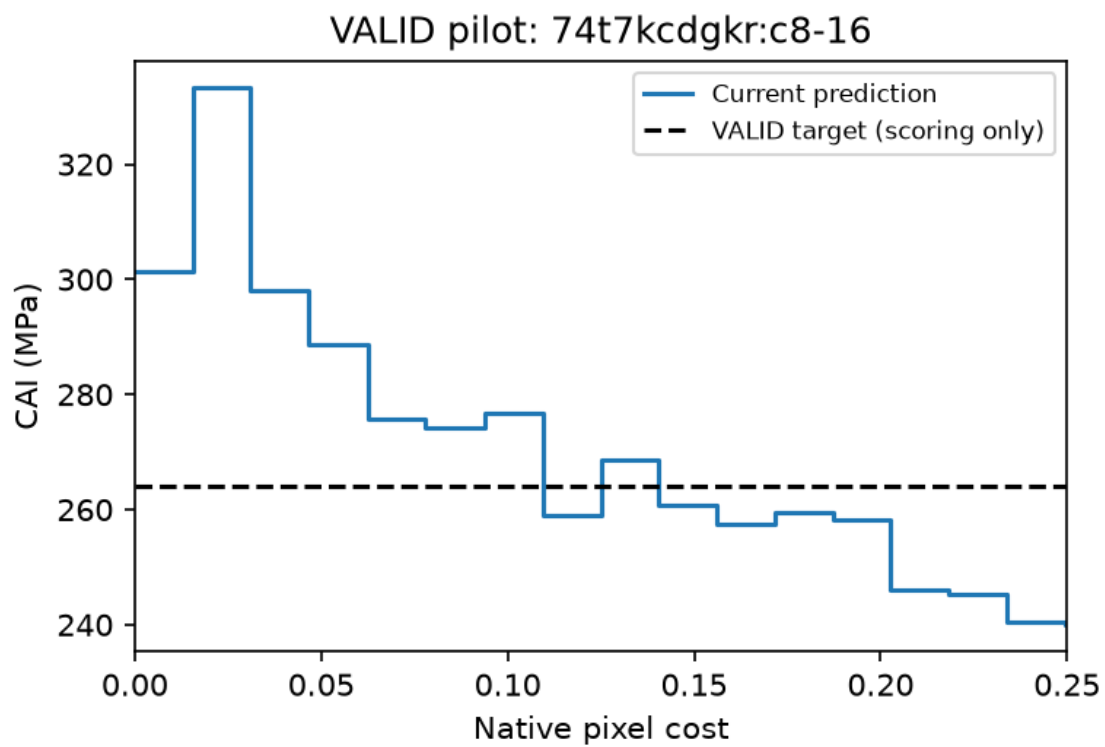


Figure S7: 74t7kcdgkr: c8-16 — prediction. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

VALID surface + frozen VLM cues

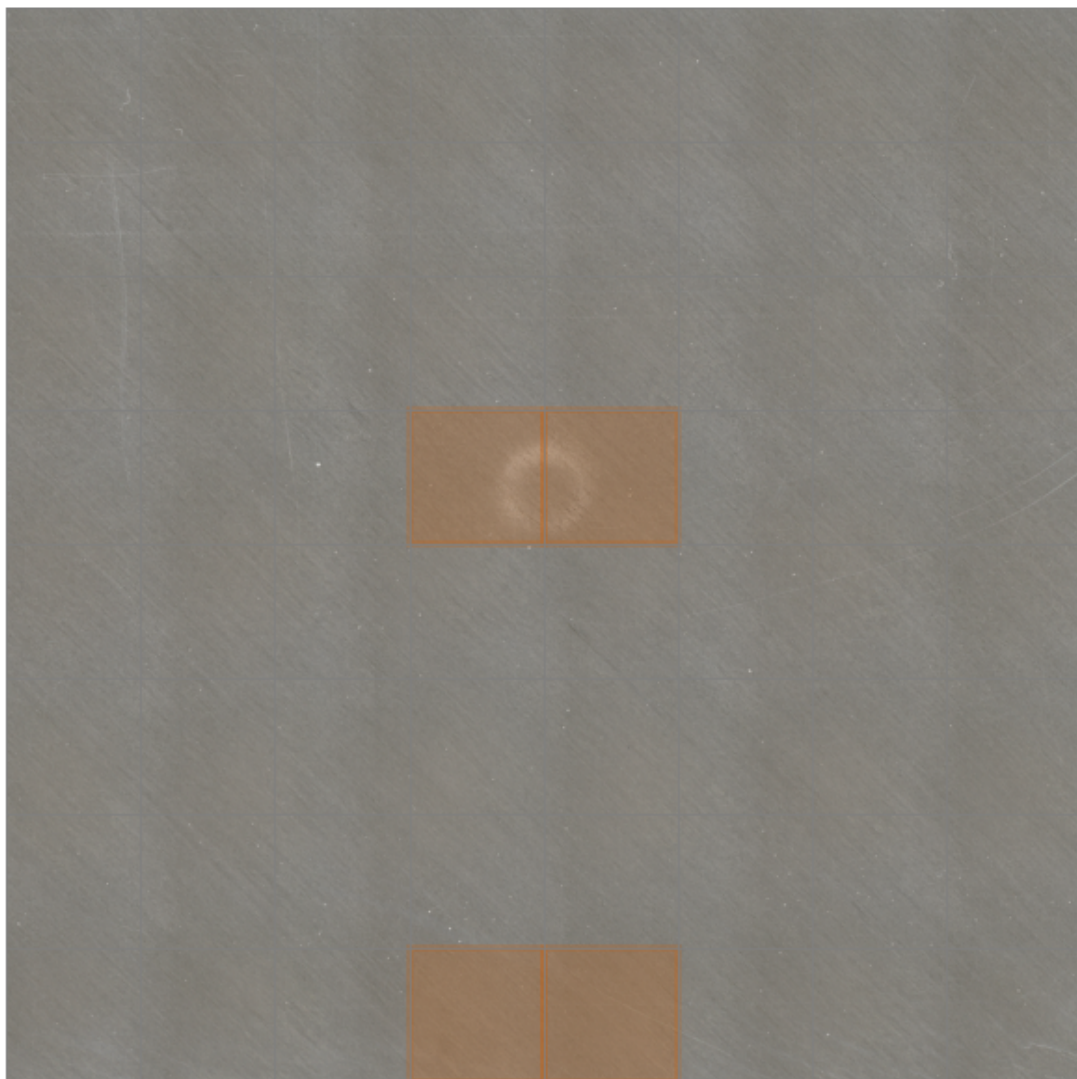


Figure S8: w68dtmpfyf: q16-29 — surface cues. Actual saved inputs, observations and predictions; unchanged source overlay.

VALID k=16, cost=0.2498
budget endpoint; gray = unobserved

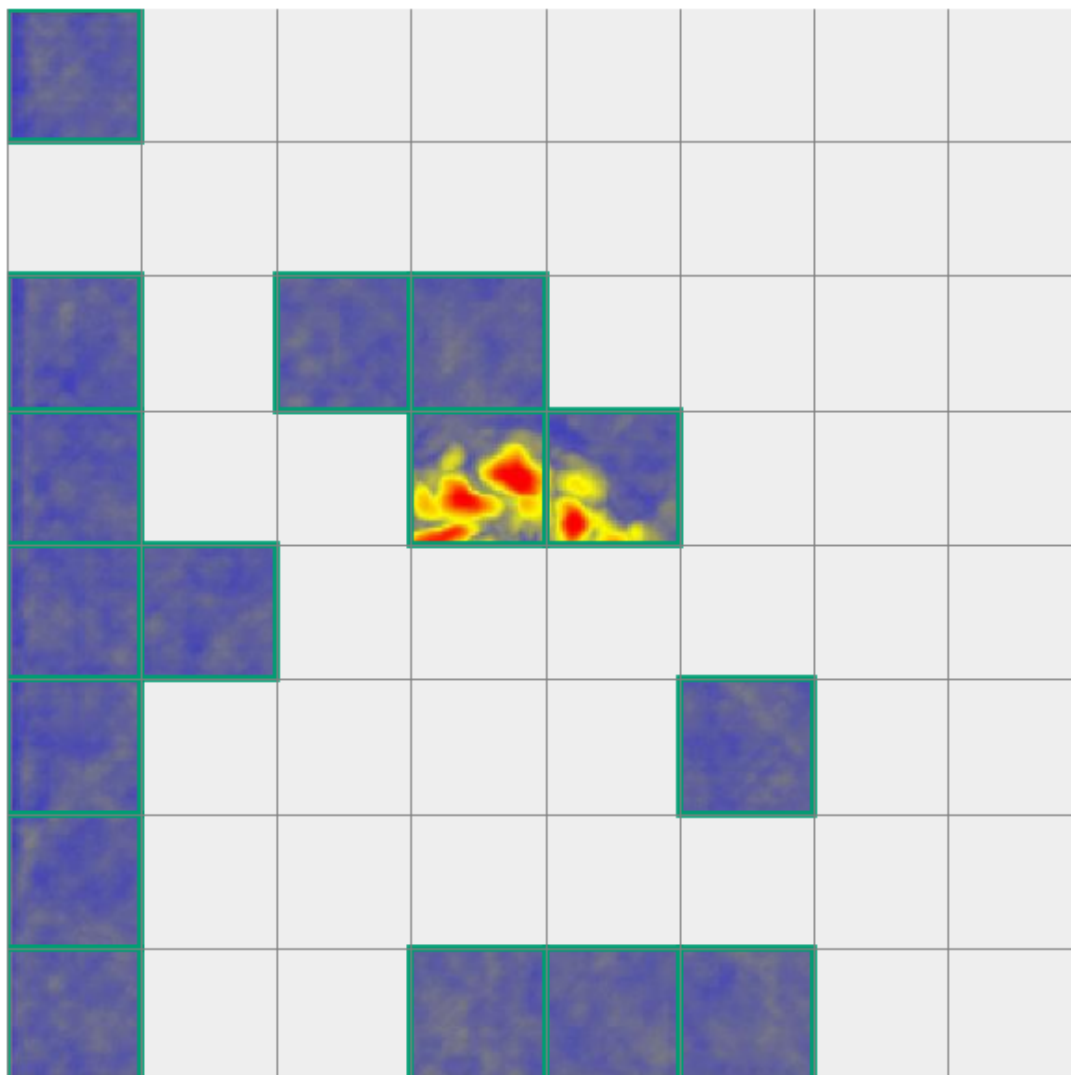


Figure S9: w68dtmpfyf: q16-29 — measured 16. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

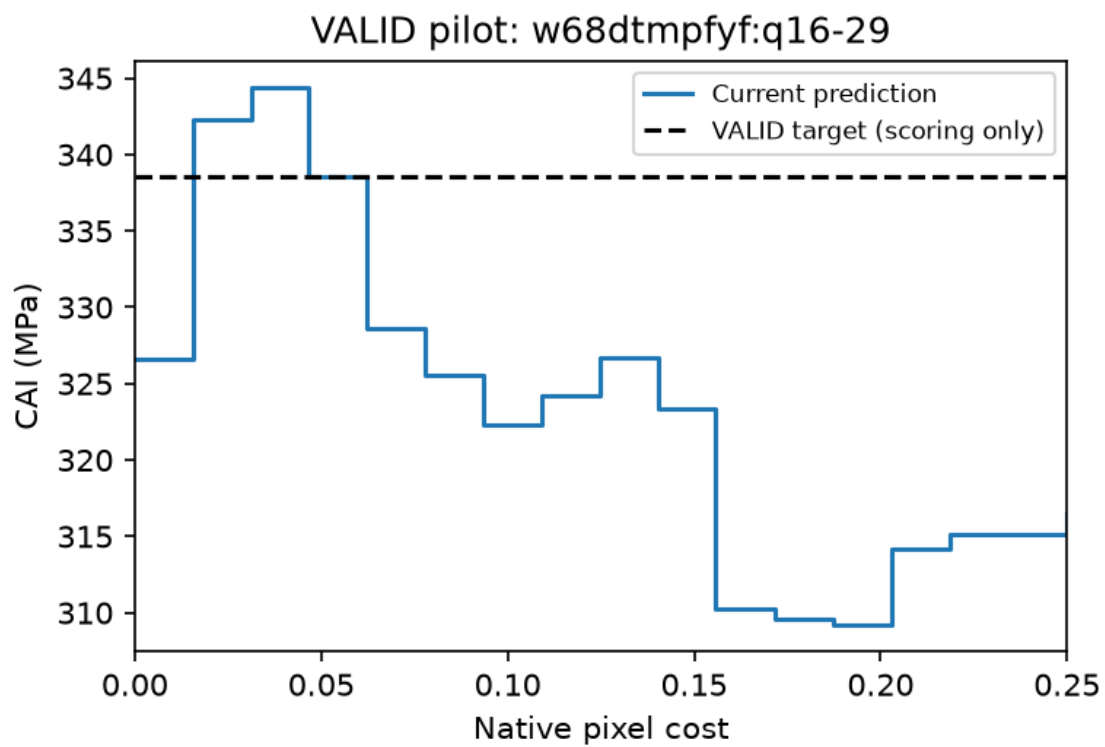


Figure S10: w68dtmpfyf: q16-29 — prediction. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

VALID surface + frozen VLM cues

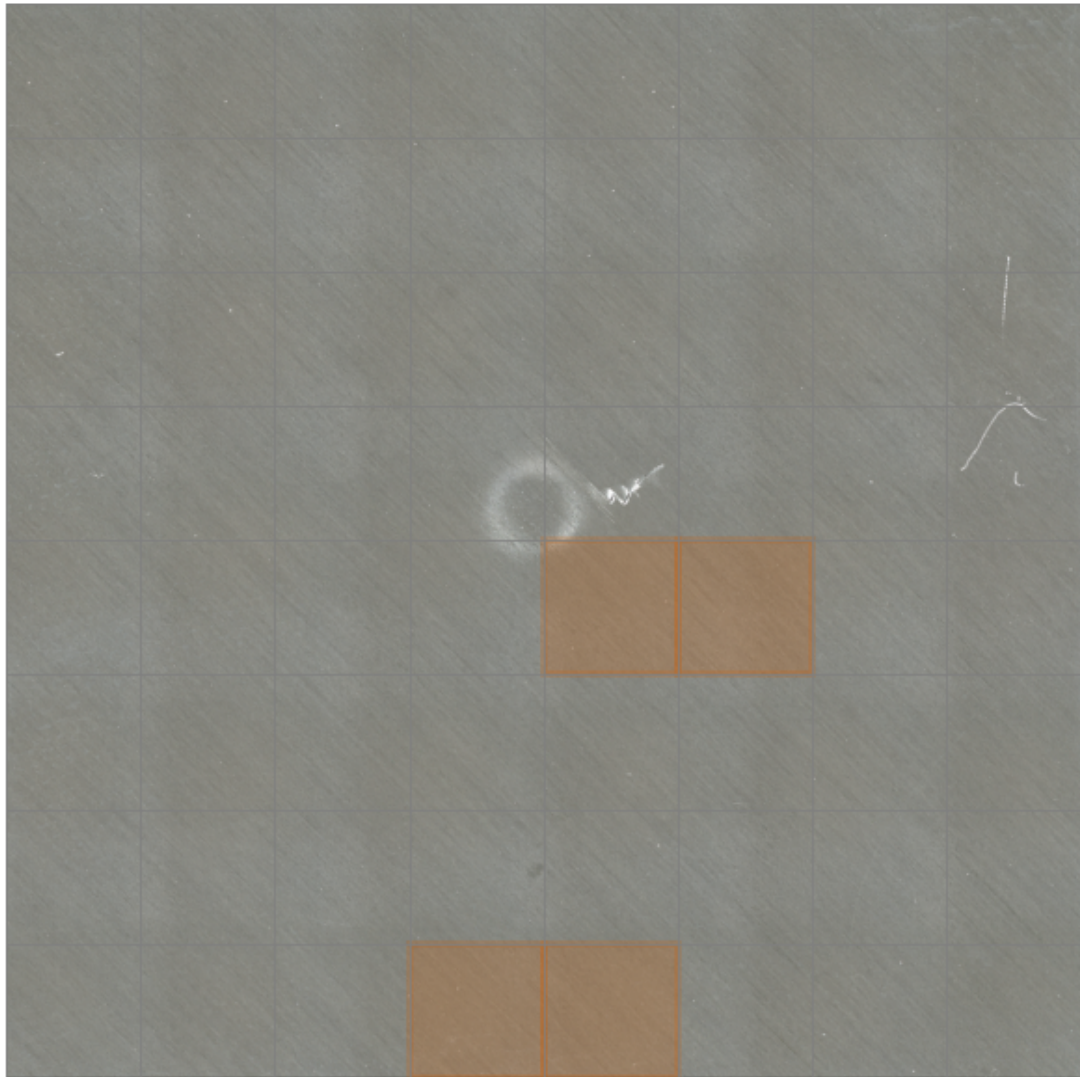


Figure S11: cgtnjyggtn: q24-48 — surface cues. Actual saved inputs, observations and predictions; unchanged source overlay.

VALID k=16, cost=0.2494
budget endpoint; gray = unobserved

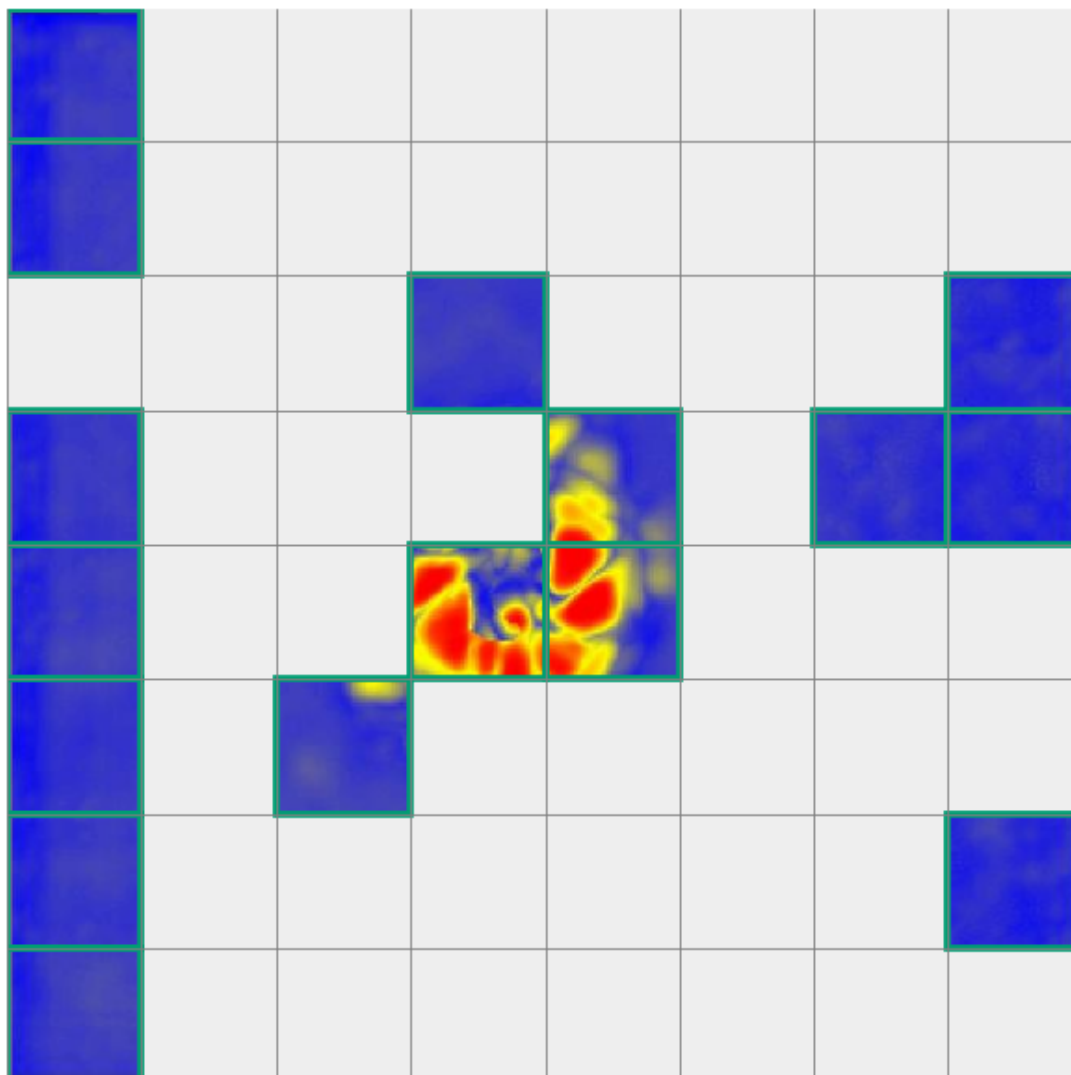


Figure S12: cgtnjyggm: q24-48 — measured 16. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

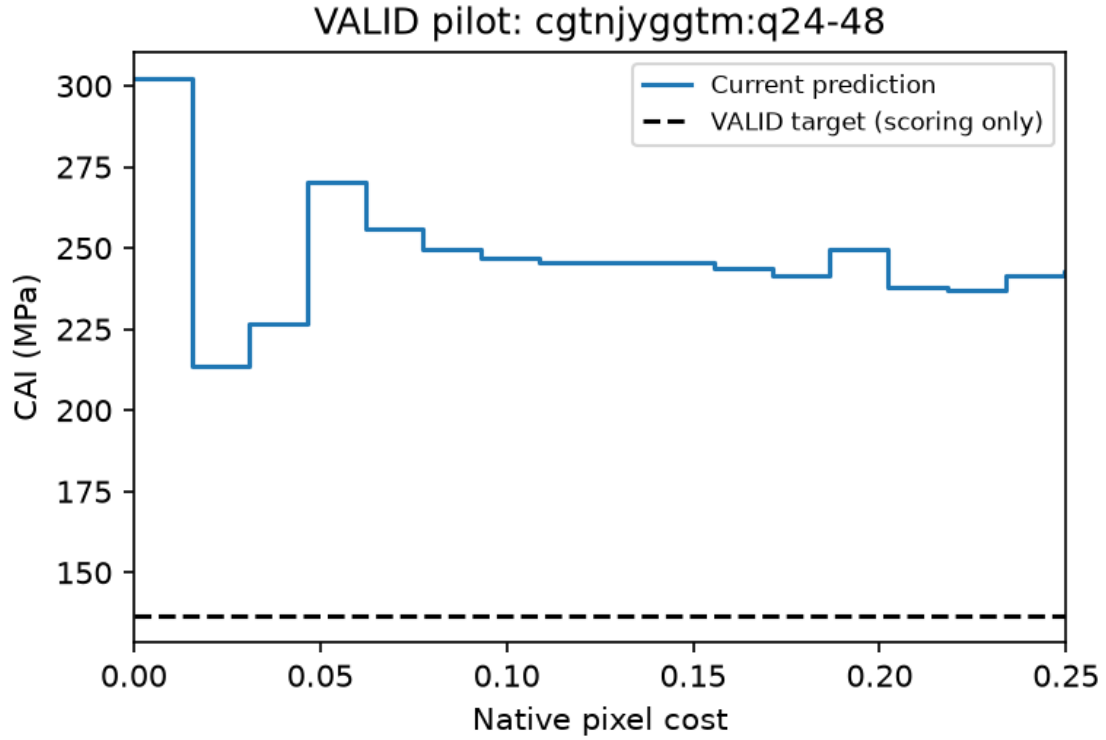


Figure S13: cgtnjygggm: q24-48 — prediction. Frozen final acquisition state or prediction trajectory; higher-error outcomes remain visible.

S6 Closest-work comparison

Work	Task endpoint	Information available to decision	Feedback and learning	Cost / termination	Verified evidence and distinction
Mack et al. (2026)	CAI strength / impact energy regression	C-scan damage image input	ResNet18 regression described; acquisition feedback unverified	Acquisition-cost protocol unverified	Institutional abstract only; predictor input established, no module-level absence claim
Fuentes et al. (2020)	Damage indication / component damage probability	Measurements collected at selected locations	Sequential novelty-index field and Bayesian optimization	Reduces observation count; exact termination unverified	Institutional abstract only; differs in endpoint, not a reproduced baseline
Shim et al. (2018)	Cost-sensitive classification	Acquired feature set	Joint classifier/acquirer with set encoding	Acquire or stop/predict; specified feature costs	PDF pp1–4, Section 3; differs from fixed-cap frozen-regressor evaluation
Janisch et al. (2019)	Costly-feature classification	Currently acquired features	Deep RL; optional external classifier described	Feature requests or classification actions	Author PDF pp1–3, Problem definition; no claim of CAI evaluation

Work	Task endpoint	Information available to decision	Feedback and learning	Cost / termination	Verified evidence and distinction
Covert et al. (2023)	Dynamic predictive feature selection	Observed feature subset	Greedy CMI / amortized optimization	Fixed feature count, uniform costs in formulation	PDF Sec2–4; squared-loss regression conditions do not establish this loss’s optimality
This study	CAI trajectory and endpoint error	Surface, numerical prior, acquired internal descriptors	Frozen regressor with learned acquisition actor	Native-pixel cap; whole-cell affordability termination	Local model/policy/metric code and saved selected-VALID trajectories

No source is marked FULLTEXT_READ when only its abstract or selected sections were read. Three algorithmic neighbours were read at the sections supporting the comparisons; unavailable engineering-neighbour details remain unverified. No cross-study accuracy leaderboard is constructed.

S7 Source-data index

File in tables/	Rows	Status
same_cost_metrics.csv	46	Unchanged frozen source
same_cost_paired_summary.csv	40	Unchanged frozen source
same_cost_metrics_by_domain.csv	270	Unchanged frozen source
mechanism_effects.csv	8	Unchanged frozen source
mechanism_area_intervals.csv	4	Unchanged frozen source
equal_quality_grid.csv	420	Unchanged frozen source
equal_quality_anchors.csv	420	Unchanged frozen source
quality_targets.csv	42	Unchanged frozen source
full_scan_gap.csv	45	Unchanged frozen source
full_scan_reference.json	JSON	Unchanged frozen source
method_summary.csv	9	Unchanged frozen source
case_same_cost_states.csv	195	Unchanged frozen source
case_figure_reuse.csv	21	Unchanged frozen source

The analysis/ directory supplies timing_contributions.csv (36 rows), timing_identity_checks.json, timing_analysis.py and six prescribed numerical checks. No original event, model or feature cache is duplicated in this manuscript package.

References

- [1] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Dataset for surface and internal damage after impact on CFRP laminates. *Data in Brief*, 43:108462, 2022. doi: 10.1016/j.dib.2022.108462. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC9294053/>.
- [2] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Dataset of compression after impact testing on carbon fiber reinforced plastic laminates. *Data in Brief*, 60:111509,

2025. doi: 10.1016/j.dib.2025.111509. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC11999467/>.
- [3] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 8-layer cross-ply laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/74t7kcdgkr/1>.
 - [4] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 16-layer cross-ply laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/yfxyg8jm46/1>.
 - [5] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 24-layer cross-ply laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/xcmzfsbd9t/1>.
 - [6] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 8-layer quasi-isotropic laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/ykhs7s2dck/1>.
 - [7] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 16-layer quasi-isotropic laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/w68dtmpfyf/1>.
 - [8] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens after low-velocity impact tests: 24-layer quasi-isotropic laminates. Mendeley Data, 2022. URL <https://data.mendeley.com/datasets/cgtnjyggtm/1>.
 - [9] Saki Hasebe, Ryo Higuchi, Tomohiro Yokozeki, and Shin-ichi Takeda. Datasets on CFRP specimens subjected to compression after impact tests. Mendeley Data, 2025. URL <https://data.mendeley.com/datasets/8scdmfdcfb/3>.
 - [10] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL Technical Report. *arXiv*, 2025. doi: 10.48550/arXiv.2502.13923. URL <https://arxiv.org/abs/2502.13923>.
 - [11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. *arXiv*, 2015. doi: 10.48550/arXiv.1512.03385. URL <https://arxiv.org/abs/1512.03385>.